

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads	)	
to the Interstate Transmission System	)	Docket No. RM26-4-000
	)	
	)	

REPLY COMMENTS OF GOOGLE LLC

Google LLC (“Google”) hereby submits these reply comments in connection with the Advanced Notice of Proposed Rulemaking proposal (“DOE ANOPR Proposal”) in the above-captioned proceeding. This proceeding has, for good reason, produced an already extensive record, including 180 initial comments that collectively stretch over 2,500 pages. While the parties may disagree on many aspects of *how* to achieve the principal goal laid out in Secretary Wright’s letter—ensuring the “timely, orderly, and non-discriminatory” interconnection of large loads to the transmission system—they overwhelmingly agree on the importance of that goal to our power system, our economic growth, and our national security.

The question now before the Commission is what conclusions to draw from that extensive record and what actions it should take to ensure “timely, orderly, and non-discriminatory” interconnection of large loads to the transmission system. Google continues to believe that the challenge of interconnecting large loads to the transmission system is, at its core, an energy infrastructure challenge.¹ The reality is that our nation’s electric transmission infrastructure is

¹ See Google Initial Comments at 2-5.

failing to keep pace with the massive, concurrent demands being driven by electrification, the resurgence of U.S. manufacturing, and the expansion of digital infrastructure. Inadequate electric transmission infrastructure is becoming the principal limitation to interconnecting new large loads, including data centers, and the needed generation capacity at the speed and scale necessary to sustain our global economic competitiveness. As explained in Google’s initial comments, the combination of aging infrastructure, decades of underinvestment, and fragmented, highly localized planning practices have resulted in a transmission grid that is struggling to keep up with the demands of this new growth, including from data centers.² As a result, it takes too long to interconnect new large loads and new generation, and the cost of doing so is too high. Lengthy interconnection delays are directly driving up energy prices. When new supply remains locked in a queue, it cannot meet rising demand in time, which puts upward pressure on all prices, including for energy, capacity, and ancillary services. Resolving those bottlenecks by building the necessary transmission infrastructure is one of the principal steps our nation can take to solve all of those challenges.

The record assembled in this docket so far corroborates that conclusion. A diverse set of stakeholders have submitted a compelling evidentiary case that increasing transmission capacity is essential to expediting the interconnection of large loads to the transmission system in order to support national competitiveness, including winning the AI race.³ For example, the Data Center

² Google Initial Comments at 3.

³ See ACEG Comments at 2; ACORE Comments at 2-5; AEP Comments at 2, 15-24; CAISO Comments at 4; DCC Comments at 8-10; Industrial Customer Organizations Comments at 3-4, 11; Joint Load Flexibility Parties Comments at 2; Microsoft Comments at 1; MISO Comments at 8; National Grid

Coalition (“DCC”) agrees that load interconnection reform is not a procedural endeavor, but “fundamentally an infrastructure challenge.”⁴ DCC points out that the grid as we know it today was designed “to accommodate incremental changes in peak demand”—not to keep pace with the unprecedented growth we are seeing today.⁵ Under existing reactive and siloed planning approaches, “transmission planners study new load or generation interconnection requests individually and identify incremental transmission upgrades necessary to support those requests.”⁶ The result is “a hodgepodge development of the transmission system”⁷ and plans that “only respond[] to present—or even past—reliability needs.”⁸ This piecemeal approach to transmission infrastructure planning “will not deliver infrastructure on the timeline needed” to meet the present needs of large load customers⁹ because “an approach which focuses solely on interconnecting large loads to the electric grid will not fully support rapid data center and industrial load growth.”¹⁰

Comments at 2; SEIA Comments at 22; *see also* Electricity Customer Alliance Comments at 2-3 (advocating for improved transmission planning) PJM Comments at 2 (same).

⁴ DCC Comments at 8. DCC points out that this framing is grounded in precedent; the Commission has long recognized the inextricable link between transmission access and its charge under the FPA. *Id.* at 8-9 (citing Order Nos. 888, 2000, 1920, and 1920-A).

⁵ *Id.* at 9.

⁶ AEP Comments at 15.

⁷ *Id.*

⁸ *Id.* at 18.

See AEP Comments at 15; *see also* DCC Comments at 8 (“[T]he grid was not expanded or modernized at the pace that changing system needs require.”).

¹⁰ AEP Comments at 2. Faced with extraordinary load growth in recent years, AEP observes that the transmission grid is “at the crux of this issue.” *Id.* at 21. AEP suggests several innovative interventions to meet this challenge, like proactive, scenario-based transmission planning informed by congestion data and resource forecasts, and cost allocation approaches that enable the construction of extra high voltage transmission facilities with broad systemwide benefits. *Id.* at 15-24.

Comments from grid operators,¹¹ transmission owners,¹² large load customers,¹³ and other concerned parties¹⁴ echo similar exhortations to make transmission planning processes more efficient and holistic, thereby reducing the barriers to building transmission infrastructure. The record is clear: to get large load interconnection right, the Commission must consider the interplay between transmission capacity and the interconnection of both large loads and the generators that supply them.

The way to address that foundational transmission issue is to adopt a holistic transmission planning process that optimizes grid development by accounting for both generators *and* the large loads planning to interconnect. Doing so will enable the United States to develop the most efficient, cost-effective electric transmission backbone necessary to rapidly interconnect all loads, including data centers, as well as the generation resources needed to provide them with reliable

¹¹ See CAISO Comments at 4 (noting that large loads require “more generating capacity and transmission capacity to meet demand”); MISO Comments at 8 (emphasizing the “role a robust transmission system plays in ensuring a resilient and cost-effective foundation to meet this new load growth”); PJM Comments at 2 (encouraging near-term regulatory guidance on transmission planning).

¹² See AEP Comments at 15-24; National Grid Comments at 2 (calling for rules that facilitate the necessary investments in transmission infrastructure); *see also* WIRES Comments at 8 (highlighting WIRES’ members’ readiness to deploy transmission infrastructure to meet rising demand).

¹³ See Industrial Customer Organizations Comments at 3-4, 11 (acknowledging the “substantial and immediate demand” for transmission infrastructure driven by rapid deployment of large loads and encouraging the Commission to focus near-term efforts on optimizing existing transmission system capacity); Microsoft Comments at 1 (noting that expansion of AI data center operations “necessitates significant investment in electric transmission and generation capacity”).

¹⁴ See ACEG Comments at 2 (noting that achieving U.S. energy dominance and AI leadership requires “significant expansion of the grid”); ACORE Comments at 2-5 (arguing that integrating large loads will require a modernized and expanded grid); American Conservation Coalition Comments at 4 (describing expanded transmission as the “backbone” of a reliable and affordable grid); Electricity Customer Alliance Comments at 2 (describing transmission planning as one of three “pillars” of ensuring reliable and affordable electricity in the face of load growth); Joint Load Flexibility Parties Comments at 2 (noting that transmission will be part of the solution); SEIA Comments at 22 (advocating for a final rule that provides regulatory certainty for the timely development of transmission infrastructure).

power. And time is of the essence: Absent a change of course, electric infrastructure appears to be the one pillar of our economy and our global competitiveness where we are increasingly falling behind,¹⁵ a result that would have devastating consequences for our economic growth and national security. To avoid that result, the Commission should continue to support efforts to improve long-term regional transmission planning processes,¹⁶ and reject measures that would complicate the buildout of transmission facilities being developed as a result of those processes.¹⁷

But even the best, most efficient transmission planning takes time, and the process of planning, permitting, and building the necessary transmission infrastructure takes even longer. That is time that we don't have. To ensure our national competitiveness, the United States must focus on expediting the development and interconnection of large loads, including data centers, that maximize the use of existing transmission infrastructure and minimize strain on the grid. That is the only way to build digital infrastructure at the "speed of national need"¹⁸ in the near-term while we *also* develop the transmission infrastructure that will support our long-term economic

¹⁵ For example, between 2019 and 2024, China's total electricity output grew by 36%. During the same interval, U.S. total electricity output only grew by 5%. Gavin MaGuire, *China's power grid clean-up puts several US systems to shame*, Reuters (Mar. 7, 2025), <https://www.reuters.com/markets/commodities/chinas-power-grid-clean-up-puts-several-us-systems-shame-maguire-2025-03-07/>.

¹⁶ See, e.g., *Bldg. for the Future Through Elec. Reg'l Transmission Plan. & Cost Allocation*, Order No. 1920, 187 FERC ¶ 61,068, *order on reh'g & clarification*, Order No. 1920-A, 189 FERC ¶ 61,126 (2024), *order on reh'g & clarification*, Order No. 1920-B, 191 FERC ¶ 61,026 (2025).

¹⁷ See, e.g., *N.D. Pub. Serv. Comm'n et al. v. Midcontinent Indep. Sys. Operator Inc.*, Complaint of the Concerned Commissions and Request for Expedited Action and Fast Track Processing, Docket No. EL25-109-000 (filed July 30, 2025).

¹⁸ *President's Budget Request for the U.S. Dep't of Energy for Fiscal Year 2026: Hearing Before the S. Comm. on Energy & Nat. Res.*, 119th Cong. 5 (June 18, 2025) (statement of Christopher Wright, Secretary, U.S. Dep't of Energy), <https://www.energy.senate.gov/services/files/35D195A3-946E-46D0-A5E7-84C06F46B9A4>.

ambitions. Accordingly, Google submits that this proceeding should also develop the rules and regulations that will facilitate and incentivize investments in both digital and energy infrastructure that maximize the utilization of the existing transmission system, allowing loads to start interconnecting now.

As discussed further below, the record compiled to date in this proceeding identifies and explains multiple different ways that the Commission can achieve that goal. Google urges the Commission to act promptly to facilitate those pathways to efficient development of digital and energy infrastructure while at the same time not losing focus on the transmission development imperative that is necessary to achieve durable, lasting economic competitiveness of the United States.

I. The Commission Should Focus on Maximizing Use of Existing Infrastructure and Minimizing Strain on the Grid

The record in this proceeding provides strong support for several approaches to maximize the use of the existing transmission system and incentivize the investments in digital and energy infrastructure that do the same. Those include the co-location of load and generation behind the same point of interconnection,¹⁹ the integration of generators that are electrically proximate to new

¹⁹ See AI Supply Chain Alliance Comments at 2; Amazon Comments at 6-7; American Conservation Coalition Comments at 2; America's Power Comments at 4-5; CEBA Comments at 13-14, 16-17; Constellation Comments at 2, 14-15, 38; DTE Electric Comments at 8-11; Illinois AG Comments at 8; Maryland PSC Comments at 6-7; Meta Comments at 6; Microsoft Comments at 8-9; OMS Comments at 13; PPL Companies Comments at 2, 16-20; PSEG Comments at 14.

loads,²⁰ and the interconnection of loads that can operate flexibly.²¹ Google urges the Commission to focus on developing incentives²² to promote all three avenues as the next stage in this proceeding, ensuring that any expedited processes are technology neutral and do not arbitrarily favor loads paired with generation over those capable of providing equivalent relief through flexibility.

A. Expediting Co-Location to Minimize the Need for New Transmission Infrastructure

There is broad agreement in the record that the co-location of loads and generation behind the same point of interconnection is a promising vehicle to make the most out of the existing infrastructure we have.²³ As Constellation explains, “[o]ne of the primary attributes of co-location configurations (all of them) is that they take advantage of existing infrastructure and

²⁰ See Advanced Energy United Comments at 14-15, 22-23; Base Power Comments at 2, 5-7; DCC Comments at 5; EEI Comments at 18; Eolian Comments at 18; Illinois AG Comments at 8; Maryland PSC Comments at 6-7; Ohio FEA Comments at 5; PPL Companies Comments at 2, 18; SPP TOs Comments at 5.

²¹ See American Conservation Coalition Comments at 2; CAISO Comments at 12; Constellation Comments at 23-26; Digital Energy Council Comments at 2; DTE Electric Comments at 11-12; Eolian Comments at 25-33; Fervo Energy Comments at 4-5; Joint Load Flexibility Parties Comments at 2-4; Microsoft Comments at 10; OMS Comments at 13; OpenAI Comments at 2; PG&E Comments at 14-15; State Entities Comments at 8-9; Texas Blockchain Council Comments at 2; Vantage Data Centers Comments at 7-10.

²² Google Initial Comments at 6; *see also* AES Comments at 12 (supporting the DOE ANOPR Proposal’s goal of incentivizing co-location); CAISO Comments at 12 (arguing that an expedited interconnection process for curtailable loads “would appropriately incentivize more developers to consider curtailable and dispatchable facilities, which ultimately will benefit large loads, the grid, and ratepayers”); Digital Energy Council Comments at 2 (supporting incentives for flexibility); Vantage Data Centers Comments (explaining that “practical rule sets and incentives can help bolster market interest” in data center load flexibility).

²³ See Amazon Comments at 6-7; America’s Power Comments at 2-3, 4-5; Constellation Comments at 38; Joint Consumer Advocates Comments at 18; Maryland PSC Comments at 6-7; Meta Comments at 5; Microsoft Comments at 8-9; OMS Comments at 13; PPL Companies Comments at 17; PSEG Comments at 14.

therefore minimize costs.”²⁴ Chevron adds that the Commission “should ensure that the steps it takes in this proceeding prioritize the development of hybrid arrangements that involve major investments in new generation resources to achieve the maximum electric supply buildout necessary for domestic data centers as soon as practicable.”²⁵ Indeed, co-location “can dramatically reduce infrastructure costs, improve local reliability, and allow load additions without extensive system upgrades. These configurations minimize dependence on stressed transmission corridors, reduce line losses, and allow flexible, modular expansion.”²⁶

Commenters across the industry—including hyperscalers and other large load customers,²⁷

²⁴ Constellation Comments at 38.

²⁵ Chevron Comments at 4; *see also* Advanced Energy United Comments at 13 (“To the extent that a new large load and new generation or storage resource are geographically and electrically proximate, studying them together would reduce network upgrades, lower costs, and reduce the need for critical supply chain components.”); American Transmission Company Comments at 3 (“New generation located near new load will help to serve that new load.”); DCC Comments at 5 (expressing support for studying new load and new generation together); Maryland PSC Comments at 6-7 (enumerating the benefits of co-location arrangements involving new generation/storage and new loads); PPL Companies Comments at 17-18 (expressing support for co-location configurations involving new generation and load); PSEG Comments at 14 (“[T]he benefits of co-location arrangements include speed to power, efficient utilization of the transmission system and a reduction in the need for costly network upgrades.”); Vistra Comments at 32 (urging the Commission to prioritize facilitating the coordinated and expedited study of new generation co-located with new load).

²⁶ CEBA Comments at 13.

²⁷ *See* Amazon Comments at 6-7; CEBA Comments at 8; Digital Energy Council Comments at 2; Meta Comments at 5; Microsoft Comments at 8-9; Tesla Comments at 2.

transmission providers,²⁸ state entities,²⁹ and independent power producers³⁰—support studying these facilities in a coordinated and expedited manner. And a number of commenters explain that, among the many issues raised in the ANOPR, the interconnection of co-located facilities raises the fewest jurisdictional questions.³¹ Even parties who contest FERC’s general jurisdiction over load interconnection to the transmission system agree that the Commission has authority to regulate the interconnection of a generator and load behind a single point of interconnection.³² That consensus greatly reduces the likelihood of litigation and other attendant challenges to quickly implementing solutions along these lines. The ANOPR provides the roadmap to do just that. The Commission can require processes in which co-located loads and generating facilities are studied together,³³ based on the amount of injection and/or withdrawal rights they request,³⁴ subject to the installation

²⁸ See CAISO Comments at 7; DTE Electric Comments at 8; Entergy Comments at 16 (“[W]e agree that simultaneous studies of related loads and generating facilities makes good sense[.]”); ISO-NE Comments at 6-7 (explaining the co-study construct already in place in ISO-NE); SPP Comments at 5-6 (explaining its HILLGA and HILL proposals); SPP TOs Comments at 5 (“It is essential that any large load rulemaking facilitate the ability to serve new loads with grid-connected generation in a manner similar to the SPP HILL Proposal.”); Vantage Data Centers Comments at 7-8; *see also* Duke Energy Comments at 17 (supporting the concept of coordinated load and generation evaluation but arguing that it is most efficiently done in the transmission planning process).

²⁹ See Joint Consumer Advocates Comments at 18; Missouri PSC Comments at 14.

³⁰ See Arevia Power Comments at 1; Eolian Comments at 36-40; Gridstor Comments at 5-6; LS Power Comments at 7; Vistra Comments at 31-32.

³¹ See Entergy Comments at 12; Pennsylvania PUC Comments at 11; *see also* Constellation Comments at 7 (“Interconnection of hybrid facilities, in particular, provide a crystal clear case for federal authority.”).

³² See DTE Electric Comments at 17; EEI Comments at 33-34; Pennsylvania PUC Comments at 11.

³³ *Interconnection of Large Loads to the Interstate Transmission System*, Proposed Advanced Notice of Proposed Rulemaking, Docket No. RM26-4-000, at P 20 (filed Oct. 23, 2025) (“DOE ANOPR Proposal”).

³⁴ *Id.* P 22.

of system protection facilities necessary to prevent unauthorized injections or withdrawals.³⁵

The concerns that some commenters raise about co-location should not provide a reason to forgo an efficient path to interconnecting significant sources of electricity demand with relatively little dependence on the constrained transmission system. For instance, some parties argue that load and generation studies cannot be combined because transmission providers need to model each facility separately in case one trips.³⁶ But even if the facilities must be modeled separately, there is still significant value in improving the coordination of the studies. Doing so will align project timelines, which can help reduce uncertainty and increase commercial readiness. Study coordination will also help ensure that transmission providers select the most efficient transmission network upgrades to accommodate the entire interconnection instead of less efficient, piecemeal upgrades.

There is also some debate in the record as to whether network upgrades and transmission service should be designed based on gross load or net load.³⁷ As CAISO explains, identifying network upgrades to interconnect co-located facilities based on their requested injection and

³⁵ DOE ANOPR Proposal at P 23.

³⁶ *See, e.g.*, Constellation Comments at 13-14; Indicated PJM TOs Comments at 36; MISO Comments at 23-24.

³⁷ *Compare, e.g.*, Constellation Comments at 14 (supporting net load approach for transmission service to hybrid facilities), *and* Shell Comments at 5, 9 (supporting net load approach for transmission service but with rights to request service based on gross load in the future), *and* Vistra Comments at 4, 26, 33 (supporting net load approach for network upgrades and transmission service), *with* Alliant Energy Comments at 11 (supporting gross load approach for transmission service and ancillary services), *and* EEI Comments at 3, 15-16 (same), *and* Tri-State Comments at 9 (supporting gross load approach for transmission service).

withdrawal rights is beneficial for both ratepayers and developers.³⁸ Importantly, CAISO notes that providing transmission service based on withdrawal rights will “[c]orrectly incentiviz[e] large load developers to right-size their withdrawal and injection rights,” thereby “help[ing] states, transmission planners, and load-serving entities avoid unnecessary network upgrades and additional generation procurement at ratepayers’ expense.”³⁹ Google could not agree more. Determining what network upgrades are required to accommodate the actual injection and withdrawal rights requested and providing transmission service based on the actual use of the transmission system will send the proper signals to encourage developers to pursue the most efficient projects. Doing so will provide incentives to maximize the use of existing infrastructure while minimizing strain on the grid and help get as much new demand online and powered up as quickly as possible.

Finally, some parties raise concerns about harming existing generator interconnection processes that were recently revised to comply with Order No. 2023.⁴⁰ Google agrees that the Commission should do no harm to those processes, but facilitating co-located load and generation need not necessarily occur within the existing generator interconnection cluster study process, unless the transmission provider believes that doing so would be more efficient.⁴¹ In any case, the Commission has a long history of authorizing regional variation, particularly in the RTO/ISO

³⁸ CAISO Comments at 8 (supporting studies based on net injection and/or withdrawal if coupled with protection facilities).

³⁹ *Id.* at 15.

⁴⁰ *See, e.g.*, NYISO Comments at 5-6.

⁴¹ *See* ISO-NE Comments at 6 (explaining that ISO-NE already considers new large loads in its cluster study process).

regions, and there is no reason it should not continue to do so here to ensure that any effects on the existing generator interconnection queues are minimized.⁴²

The concerns raised in the record about co-location are readily resolvable, and do not outweigh its potential value in interconnecting new large loads in a manner that maximizes the use of existing infrastructure and minimizes strain on the grid. The Commission should act expeditiously to facilitate a pathway to study and interconnect co-located facilities in a coordinated and expedited manner. Nevertheless, any such pathway must be technology neutral. A Commission-established fast track for loads using co-located generation to mitigate transmission impacts should be paired with equivalent pathways to achieve the same end, including, for example, using curtailment. Such pathways must be voluntary, incentive based, and limited only to those loads that agree to meet the technical requirements of participating under the relevant frameworks and include measures to ensure that they do, in fact, comply with those requirements. The following sections provide a more detailed discussion on these additional pathways.

⁴² See, e.g., *Reg'l Transmission Orgs.*, Order No. 2000, FERC Stats. & Regs. ¶ 31,089, at 31,073 (1999) (cross-referenced at 89 FERC ¶ 61,285), *order on reh'g*, Order No. 2000-A, FERC Stats. & Regs. ¶ 31,092 (2000) (cross-referenced at 90 FERC ¶ 61,201), *aff'd sub nom. Pub. Util. Dist. No. 1 of Snohomish Cty. v. FERC*, 272 F.3d 607 (D.C. Cir. 2001) (stating there is no “one size fits all” approach to RTO governance); *Offer Caps in Mkts. Operated by Reg'l Transmission Orgs. & Indep. Sys. Operators*, Order No. 831, 157 FERC ¶ 61,115, at P 147 (2016) (permitting regional variation for treatment of incremental energy offers above \$1,000/MWh that the RTO/ISO or Marketing Monitoring Unit cannot verify prior to the start of the market clearing process); *Wholesale Competition in Regions with Org. Elec. Mkts.*, Order No. 719, 125 FERC ¶ 61,071, at P 62 (2008) (permitting regional variation for RTOs/ISOs developing requirements for demand response resources participating in ancillary service markets), *order on reh'g*, Order No. 719-A, 128 FERC ¶ 61,059, *order on reh'g*, Order No. 719-B, 129 FERC ¶ 61,252 (2009).

B. Expediting the Interconnection of Electrically Proximate Generation and Load Can Help Grid Operators Get the Most out of Existing Infrastructure

The record also provides strong support for facilitating the interconnection of generation resources paired with new large loads, not just in traditional co-location configurations, but also in electrically proximate pairs.⁴³ DCC, for instance, contends that joint study of new, physically proximate load and generation presents meaningful system benefits, which include reducing the need for network upgrades, accelerating the timeline for bringing load and generation online, improving resource adequacy, minimizing congestion and curtailment, and expanding siting options.⁴⁴ After all, “[i]t is obvious from basic engineering principles that generation located near load requires less transmission infrastructure to serve that load than generation located distant from load.”⁴⁵ Google agrees with commenters that the Commission should establish pathways for facilitating the expedited study and interconnection of electrically proximate pairs of generation

⁴³ See Advanced Energy United Comments at 14-15, 22-23; Base Power Comments at 2, 5-7; DCC Comments at 5; EEI Comments at 18; Eolian Comments at 18; Illinois AG Comments at 8; Maryland PSC Comments at 6-7; Ohio FEA Comments at 5; PPL Companies Comments at 2; SPP TOs Comments at 5. Separately, though relatedly, AEP urges the Commission to address generator interconnection queue backlogs by enabling connect-and-manage interconnection. AEP Comments at 7-10. AEP urges the Commission to look to the Electric Reliability Council of Texas’s (“ERCOT”) approach to “connect and manage,” which it contends is a good starting point. *Id.* at 8. AEP explains that, under ERCOT’s connect and manage approach, generators may interconnect to the transmission grid once they satisfy certain technical criteria, even when network constraints remain or transmission upgrades are incomplete. *Id.* According to AEP, “[a]ny grid bottlenecks caused by a new generator are managed by ERCOT through market redispatch and curtailment, until they are later addressed through broader transmission planning processes,” thereby decoupling interconnection from network expansion. *Id.*

⁴⁴ DCC Comments at 5.

⁴⁵ Eolian Comments at 18; *see also* Base Power Comments at 7 (“[E]valuating dispatchable resources “in relation to their effect on the POI, rather than by strict interconnection topology . . . ensures that operationally relevant capacity, whether physically co-located with the load or situated a few substations away, can be considered within interconnection and planning studies.”).

and load, as doing so will reduce the network upgrades needed to interconnect new resources, which can accelerate the pace at which new generation comes online.

Notably, SPP’s recently filed High Impact Large Loads Generation Addition (“HILLGA”) proposal—which has garnered support from a diverse set of stakeholders⁴⁶—offers one example of a framework for expediting the study and interconnection of new generation to serve electrically proximate large loads. The HILLGA proposal allows for a generator which supports a proximate High Impact Large Load (“HILL”) to opt into an interconnection process that allows for limited, expedited interconnection and in which the generator is studied in 90 days.⁴⁷ Generators interconnecting via the HILLGA process receive Load Limited Resource Interconnection Service (“LLRIS”), a limited and provisional form of interconnection service that allows the generator to serve a nearby HILL, rather than traditional generator interconnection service (i.e., Network Resource Interconnection Service (“NRIS”) or Energy Resource Interconnection Service (“ERIS”)).⁴⁸ After five years, LLRIS terminates and the generator must transition to NRIS or ERIS.⁴⁹

⁴⁶ See generally *Sw. Power Pool, Inc.*, Docket No. ER26-247-000.

⁴⁷ *Sw. Power Pool, Inc.*, Submission of Tariff Revisions to Add the High Impact Large Load Processes and High Impact Large Load Generation Assessment, Docket No. ER26-247-000, at 23-31 (filed Oct. 24, 2025) (“SPP HILLGA Transmittal”) (explaining that the POI for the generator interconnection must be in close electrical proximity to the HILL, which means either two substations away from the HILL or within one substation from the HILL if the generation facility is connecting to a 765 kV transmission line).

⁴⁸ SPP HILLGA Transmittal at 23, 29-30.

⁴⁹ *Id.* at 36.

The Commission can use some of the key concepts of the HILLGA proposal as a starting point to create a regime for expediting the interconnection of new generation that is electrically proximate to a new large load. Specifically, the Commission should ensure that any such regime:

- Provides an optional pathway (*i.e.*, an incentive) and not a requirement that effectively blocks the interconnection of large loads that, for whatever reason, are not or cannot be paired with electrically proximate generation.
- Is time-limited and requires the generator to transition to full interconnection service on a definite timeline (*e.g.*, the five-year limit in HILLS/HILLGA). This will help balance the competing interests of ensuring that new generators can interconnect quickly while also ensuring that, over the medium and long terms, we develop the necessary transmission infrastructure.
- Ensures that any expedited interconnection process is technology neutral. In particular, it must be open to storage resources, which can deploy more rapidly than any other grid resource and can manage the impact of large loads on the grid.⁵⁰
- Create a study timeline and milestones that are both feasible and that facilitate the rapid interconnection of new generation resources. In this vein, the Commission should require that the system impact study be completed within 90 days and create a defined, time-limited negotiation window during which parties can negotiate an interconnection agreement.
- Ensure that generators using the expedited interconnection process are electrically proximate to the load they will serve during the period that they have limited interconnection service. On this point, Google agrees that what constitutes “electrical proximity” may vary by region and that the term should be defined at a sufficient level of generality “to allow consideration of capacity deliverable to the associated load such that it relieves, avoids, or minimizes congestion impacts.”⁵¹

To maximize the impact of these reforms, the Commission should pair this expedited generator interconnection path that provides temporary, limited interconnection service for

⁵⁰ Eolian Comments at 27 (citing Joseph Rand et al., Lawrence Berkeley Nat’l Lab’y, *Queued Up: 2024 Edition: Characteristics of Power Plants Seeking Transmission Interconnection As of the End of 2023*, at 36, 40 (Apr. 2024), https://emp.lbl.gov/sites/default/files/2024-04/Queued%20Up%202024%20Edition_R2.pdf).

⁵¹ Advanced Energy United Comments at 14-15.

generation paired with electrically proximate load with the creation of a complementary transmission service. This is necessary to avoid a situation in which a newly interconnected generator that has limited interconnection service cannot deliver to the large load it is intended to serve.

C. Creating a Framework to Facilitate and Incentivize the Voluntary Participation of Curtailable Loads Can Further Enable Grid Operators to Maximize Use of Existing Grid Infrastructure

A wide and diverse set of commenters—including RTOs/ISOs,⁵² data center developers,⁵³ independent power producers,⁵⁴ states and groups representing states’ interests,⁵⁵ and utilities⁵⁶—also express strong support for incentivizing and facilitating the interconnection of curtailable⁵⁷ loads and dispatchable hybrid configurations. They argue that incentivizing load flexibility and curtailability, if done correctly, can help get the most out of the existing grid, thereby reducing the cost of network upgrades, the amount of generation needed to serve large load, the time required to interconnect large load, and customer impacts.⁵⁸ For instance, CAISO and NYISO view

⁵² See CAISO Comments at 12.

⁵³ See Digital Energy Council Comments at 2; Microsoft Comments at 10; OpenAI Comments at 2; Vantage Data Centers Comments at 7-10.

⁵⁴ See Constellation Comments at 23-26; Eolian Comments at 25-33; Fervo Energy Comments at 4-5; Joint Load Flexibility Parties Comments at 2-4.

⁵⁵ See OMS Comments at 13; State Entities Comments at 4, 8-9.

⁵⁶ See DTE Electric Comments at 11-12; PG&E Comments at 14-15.

⁵⁷ Google uses the term “curtailable” to encompass flexible and curtailable loads.

⁵⁸ See, e.g., Constellation Comments at 23 (noting that flexible loads can maximize efficient use of the transmission system, avoid system upgrades, and reduce the need to procure additional capacity); Joint Load Flexibility Parties Comments at 3 (describing load flexibility as both quicker to deploy and less expensive than new generation); Microsoft Comments at 10 (highlighting grid benefits of flexible arrangements); National Grid Comments at 19 (“Large loads that agree to curtailment or other flexibility

curtailability as an important tool for addressing reliability and resource adequacy issues exacerbated by large loads.⁵⁹ Echoing these comments, NY TOs note that relying on load flexibility can avoid reliability impacts and upgrade costs.⁶⁰ A recent study by the Nicholas Institute at Duke University, cited by many commenters, notes that a load curtailment rate of just 0.25% annually would allow for the integration of 76 GW of new load—the equivalent of 10% of the nation’s current aggregate peak demand—with minimal generation capacity expansion.⁶¹ And the State Entities explain that curtailable loads can shield ratepayers from cost risk associated with large load customers.⁶²

Google agrees with these commenters. Creating expedited interconnection pathways that incentivize loads willing to be curtailable balances the need to develop digital infrastructure with the overarching imperative of ensuring reliability and resource adequacy. Indeed, a recent study of utility transmission data co-authored by Princeton University’s ZERO Lab found that

can help reduce or defer costly network upgrades, make better use of existing transmission capacity, and accelerate interconnection timelines. These arrangements, if paired with robust operational safeguards and compliance protocols, can unlock innovation and efficiency while maintaining reliability—consistent with the objectives of the ANOPR.”); OMS Comments at 13 (noting that flexibility can “reduce the cost of network upgrades, lower the amount of generation needed to serve the interconnecting large load, shorten the time required to interconnect the prospective large load, and help shield existing customers from costs caused by the large load”).

⁵⁹ See CAISO Comments at 12 (“The ability to curtail and dispatch these facilities gives transmission providers many tools to maintain reliability that firm and non-curtailable facilities lack, especially the ability to curtail the facility in the event of a grid contingency.”); NYISO Comments at 8 (explaining that one potential “approach to alleviate [] resource adequacy needs” is “a commitment from the proposed large load project to be curtailable”).

⁶⁰ NY TOs Comments at 20.

⁶¹ Tyler H. Norris et al., *Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in US Power Systems*, at 2 (2025), <https://nicholasinstitute.duke.edu/sites/default/files/publications/rethinking-load-growth.pdf> (“Duke Nicholas Institute Study”).

⁶² State Entities Comments at 8-9.

combining flexible grid connections with a “bring-your-own-capacity” framework in PJM⁶³ could shorten interconnection wait times by three to five years and maintain grid power availability for over 99% of hours, while requiring on-site resource dispatch for only 40 to 70 hours annually.⁶⁴ The study further found that this approach allows large loads to contribute approximately \$733 million per gigawatt toward incremental costs, reducing the net system cost increase by 96% compared to inflexible scenarios.⁶⁵

Yet, despite this potential, there are few such pathways at the moment. As Constellation points out, the only network transmission service currently available “assumes full transmission service all the time.”⁶⁶ That assumption slows the speed at which grid operators can integrate large loads and increases the risk that interconnecting large loads will negatively impact existing customers.

Given that load flexibility is “economically and operationally indispensable,”⁶⁷ some regions of the country are taking steps to expedite the integration of curtailable loads. For example,

⁶³ See generally DCC et al., *Joint Proposal for CIFP Large Load Additions: Load Forecasting, Enhanced Demand Flexibility*, vBYOG (Nov. 19, 2025) <https://www.pjm.com/-/media/DotCom/committees-groups/cifp-lla/2025/20251119/20251119-item-02d---data-center-coalition-governors-exelon-and-ppl-package---executive-summary.pdf> (proposing a voluntary pathway for expedited interconnection of large loads that are matched with generation that has a one-for-one accredited capacity tied to the large load customer’s ramp and utilization schedule).

⁶⁴ Carlo Brancucci et al., Camus, encoord, and Princeton ZERO Lab, *Flexible Data Centers: A Faster, More Affordable Path to Power*, at 16 (Dec. 2025), <https://www.camus.energy/flexible-data-center-report>.

⁶⁵ *Id.* at 31.

⁶⁶ Constellation Comments at 23; see also *id.* (“[I]f new load is required to take firm service and one assumes that such load will ‘operate at 100% of its maximum electricity draw at all times, including during system-wide peaks’—which is the assumption underlying traditional NITS—then it is ‘far more likely’ that ‘significant upgrades, such as new transformers, transmission line reconductoring, circuit breakers, or other substation equipment’ will be required.” (quoting Duke Nicholas Institute Study at 5)).

⁶⁷ Lynne Kiesling Comments at 8.

SPP is developing the Conditional High Impact Large Load Service (“CHILLS”), which would create a voluntary pathway to allow curtailable large loads to expeditiously and conditionally connect to the transmission system in exchange for being curtailed during periods of system stress.⁶⁸ Under the CHILLS model, a willing flexible load could connect to the grid without designated generating assets to serve it and could eventually transition to firm transmission service.⁶⁹ Although the CHILLS proposal has not been filed with the Commission, and Google takes no position at this time whether it is just and reasonable and not unduly discriminatory or preferential, it nevertheless offers an example of how grid operators can, in fact, implement rules and solutions that harness load curtailability to increase the speed at which large loads interconnect to the grid. And while some data centers may have limited curtailment capability today, an expedited process could incentivize innovation in data center design and operations.⁷⁰

Although a limited set of commenters are opposed to the DOE’s proposal to incentivize load curtailability, their concerns, which center on operational complexity, resource adequacy and reliability, and impacts on existing customers, can be resolved while still incentivizing curtailability. For instance, the Maine Office of Public Advocate argues that an expedited interconnection process for curtailable loads would trigger price increases for other customers if

⁶⁸ *SPP’s High Impact Large Load Interconnection Solutions: Powering Growth, Accelerating Opportunity*, SPP, <https://www.spp.org/markets-operations/high-impact-large-load-hill-integration/> (last visited Dec. 3, 2025).

⁶⁹ SPP Comments at 6. SPP is also developing Price Adaptive Load Service (“PALS”), a flexible, non-firm transmission service option that allows loads to vary their participation based on the price of energy. *Id.* at 6-7.

⁷⁰ *See, e.g.*, CEBA Comments at 14-15.

new loads are not matched with new generation.⁷¹ But as the State Entities, among others, note, incentivizing curtailability can actually protect ratepayers by avoiding impacts to regional peak loads.⁷²

Other parties raise concerns regarding the resource adequacy and reliability impacts of curtailable loads.⁷³ But curtailability directly addresses those concerns. For example, according to NYISO and CAISO—two entities responsible for ensuring transmission system reliability—curtailability would be an important tool for addressing the reliability and resource adequacy impacts of large loads.⁷⁴ Finally, while concerns about the operational complexities of integrating curtailable loads would need to be addressed, that complexity is not, in and of itself, a reason to ignore the benefits that curtailability can provide. Instead, it is a reason why this Commission should focus this proceeding on facilitating pathways by which grid operators can reliably integrate and manage large curtailable loads, unlocking the benefits that they can provide for all customers.

While some regional variation may be necessary to accommodate the characteristics of different transmission systems, the Commission can articulate a set of core principles and

⁷¹ Maine OPA Comments at 4.

⁷² State Entities Comments at 4; *see also, e.g.*, Lynne Kiesling Comments at 16 (“Flexible loads that avoid peak demand (including based on grid operator signals) do not cause peak-driven upgrades or capacity additions . . .”).

⁷³ *See* Joint Consumer Advocates Comments at 24 (arguing that large loads and hybrid facilities should be able to interconnect to the system only if they bring sufficient capacity to offset the additional load brought onto the grid).

⁷⁴ CAISO Comments at 12 (noting reliability benefits); NYISO Comments at 8-9 (noting resource adequacy benefits); *see also* Oklahoma Corporation Commission and AG Comments (pointing to the CHILLS proposal as one of several SPP initiatives that, together, “represent significant progress toward improving regional planning, accommodating unprecedented load growth, and ensuring the SPP system continues to operate reliably and efficiently”).

requirements to govern the interconnection of and service to curtailable loads.⁷⁵ Those should include the following:

- Rules governing the interconnection and operation of curtailable loads must account for how large loads, including data centers, operate in practice. Such rules must encourage innovation without being so onerous and/or impractical that they discourage large loads from opting to operate flexibly when doing so could be beneficial for the load and the system.
- Reliance on curtailable loads must not stifle or interfere with the medium- and long-term buildout of transmission infrastructure for loads seeking long-term firm service. The Commission must ensure that there are pathways by which large loads can transition from curtailable to firm service, if desired.
- The process for studying and interconnecting curtailable loads must be designed to ensure that it does not inadvertently delay or interfere with existing interconnection processes.
- There must also be a set of minimum standards to ensure curtailable loads actually curtail when called on by system operators.
- Any program for integrating curtailable loads must comply with the appropriate reliability standards.
- Large loads leveraging interim curtailable service and seeking to transition to firm service must be adequately included in transmission planning models, such that grid operators develop transmission solutions to provide these loads with firm service.⁷⁶
- As part of a longer-term solution, Google reiterates the Commission should consider encouraging grid operators to adopt planning frameworks that integrate transmission, load,

⁷⁵ Order No. 841 is an example of how the Commission can develop a participation model that accommodates the needs of a resource like curtailable loads. *See Elec. Storage Participation in Mkts. Operated by Reg'l Transmission Orgs. & Indep. Sys. Operators*, Order No. 841, 162 FERC ¶ 61,127 (2018) (establishing a participation model consisting of market rules that, recognizing the physical and operational characteristics of electric storage resources, facilitates their participation in the RTO/ISO markets), *order on reh'g*, Order No. 841-A, 167 FERC ¶ 61,154 (2019), *aff'd sub nom. Nat'l Ass'n of Regul. Util. Comm'rs v. FERC*, 964 F.3d 1177 (D.C. Cir. 2020).

⁷⁶ This would address PJM's concern that, if voluntary curtailable load is not planned for as firm load in regional transmission planning processes, it could lead to a less robust buildout of the grid. PJM Comments at 10.

and interconnection planning (akin to SPP’s Consolidated Planning Process proposal),⁷⁷ which could help grid operators better plan for the future, thereby minimizing interconnection delays and costs.

II. Jurisdiction

Even the most effective proposals for facilitating large load interconnection are beside the point of this proceeding if the Commission lacks the jurisdiction or authority to implement or approve them. With that in mind, Google provides the following perspective on the jurisdictional issues raised by Secretary Wright’s letter.

Although Google agrees that the Commission has jurisdiction over the actual interconnection of large loads to the transmission system, it is important not to overstate the practical impact of that conclusion in the broader context of large load integration issues. Because states have exclusive jurisdiction over the rates, terms, and conditions of retail sales, every large load interconnection in the country will require active state involvement. The Commission affirmed that critical conclusion just weeks ago.⁷⁸ And, as other commenters have recognized, the vast majority of the contents of recent large load tariffs enacted in many states around the country would not be affected, much less preempted, if the Commission were to conclude that the actual interconnection of large loads to the transmission falls within the Commission’s jurisdiction.⁷⁹

⁷⁷ *Sw. Power Pool, Inc.*, Submission of Revisions to the SPP Open Access Transmission Tariff to Implement the Consolidated Planning Process, Docket No. ER26-414-000 (filed Nov. 3, 2025).

⁷⁸ *Cf. Tri-State Generation and Transmission Association, Inc.*, 193 FERC ¶ 61,070, at P 45 (2025) (“*Tri-State*”) (rejecting proposed large load tariff that appeared to present an impermissible intrusion on retail rate regulation where Tri-State failed to demonstrate that its proposal would not regulate the terms and conditions of a large load customer’s retail service, including the terms of energy sales from a member utility to a retail customer, minimum load and energy requirements for retail purchases by the large load customer, and monthly caps for retail energy or demand)

⁷⁹ *See* Vistra Comments at 15.

It is for that reason that Google agrees *both* with Secretary Wright’s interpretation of the Federal Power Act and with the recent NARUC resolution urging the Commission

[T]o preserve and affirm states’ retail regulatory authority under the Federal Power Act, ensure that large load interconnections do not compromise grid reliability or impose undue costs on retail customers, and respect state tools for promoting system flexibility and equitable cost allocation.⁸⁰

The visions articulated in Secretary Wright’s letter and NARUC’s resolution are eminently compatible. Google believes that this proceeding will not be successful in ensuring the “timely, orderly, and non-discriminatory” interconnection of large loads to the transmission system unless *both* visions are fully realized.

In addition, Google understands the potential reforms contemplated in the opening comments to this proceeding to exist on a spectrum in terms of the jurisdictional questions that they pose. On the one hand, the interconnection of co-located generation and large loads behind a single point of interconnection raises the fewest jurisdictional questions, as it is already firmly established that the generator’s interconnection agreement is within the Commission’s exclusive jurisdiction, and the Commission does not lose jurisdiction by virtue of the fact that a retail sale also takes place behind that point of interconnection.⁸¹ Similarly, because the Commission has exclusive jurisdiction over the rates, terms, and conditions of transmission service, and the

⁸⁰ NARUC Board of Directors, *EL-1 Resolution Urging the Federal Energy Regulatory Commission to Preserve and Affirm State Retail Regulatory Jurisdiction in its Large Load Interconnection Proceeding* (Nov. 11, 2025), <https://pubs.naruc.org/pub/2C526A94-D533-BE0A-336A-178A366C7A91>.

⁸¹ See, e.g., *PJM Interconnection, L.L.C.*, 190 FERC ¶ 61,115, at P 72 (2025) (“PJM Show Cause Order”) (explaining that the Commission has jurisdiction over interconnections “based on the nature of interconnection to the transmission system generally, not the purpose for such interconnection (i.e., to make a wholesale sale or to make a wholesale or retail purchase)”); see also Constellation Comments at 7-9.

practices directly affecting such rates, terms, and conditions, the provision of flexible, curtailable, or similar forms of transmission service as an *incentive* to facilitate the expedited interconnection of large loads to the transmission system is similarly clearly within Commission jurisdiction. The same goes for pairings of new load and new generation—sometimes called a “bring your own generation” model—especially where the incentive for such pairings is expedited interconnection of the generation resource(s).

At the other end of the spectrum, the interconnection of new loads on their own, independent of any of the variations discussed in the previous paragraph, arguably raises the most jurisdictional questions, as the Commission “has not historically exercised jurisdiction over any and all transmission-level interconnections of retail load.”⁸² As such, asserting jurisdiction over “any and all” interconnections of retail load to the transmission system would represent the most sweeping jurisdictional theory that the Commission could plausibly adopt in this proceeding.

The substantive reforms discussed above that Google is advocating for in this pleading⁸³ fall on the relatively uncontroversial end of the spectrum. Accordingly, those reforms not only minimize strain on the existing grid, but they also minimize jurisdictional friction between the Commission and the states, which Google believes will further help ensure their effectiveness, particularly in the near term.

⁸² PJM Show Cause Order, 190 FERC ¶ 61,115, at P 72 & n.224.

⁸³ *See supra* §§ I.A, I.B, I.C.

III. Conclusion

The Commission has before it a series of important and critical questions as well as a massive record on how it should answer those questions. Google submits that the Commission should focus on measures to expedite the development and interconnection of large loads, including data centers, that maximize the use of existing transmission infrastructure and minimize strain on the grid. Doing so now will do the most to protect all customers, promote our economic competitiveness, including with respect to AI, and carry out the Commission's statutory responsibilities. There is no time to waste.

Respectfully submitted,

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Dated: December 5, 2025

CERTIFICATE OF SERVICE

I hereby certify that I have this day served the foregoing document upon each person designated on the official service lists compiled by the Secretary of the Commission in these proceedings.

Dated: December 5, 2025

Washington, D.C.

/s/ Kelly Cook

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